1. Affective Mirroring in Video Game NPCs (FDG 2025)

Source
Šinković, Gómez-Maureira, Gerhold (2025). *Affective Mirroring in Video Game NPCs: A Pilot Study Evaluating Player Engagement*. FDG '25. University of Twente Research

Focus of the Paper

Explores whether **AI-driven affective mirroring**—NPCs matching a player's facial expression—can deepen attachment and engagement in a visual novel–style dating game prototype. FER is used to read the player's facial emotion and the NPC reacts with congruent emotional feedback. University of Twente Research

Study Setup

- Visual novel dating game with an emotionally responsive NPC.
- FER pipeline detects player emotion from facial expressions.
- NPCs mirror or align their reactions with the detected emotion.
- Small pilot: N = 10 participants, ages 21–24, mix of gamers and non-gamers. University of Twente Research

Key Findings

- Affective mirroring *can* increase **player attachment** to NPCs, but only when responses are:
    - **Subtle** (not cartoonishly overreactive)
    - **Context-aware** (mirroring fits the narrative moment)
- Overly literal or blunt mirroring risks breaking immersion and making the NPC feel mechanical. University of Twente Research

Design Implications for Emotion-Responsive NPCs

- Mirroring shouldn't just parrot the detected emotion; it needs a **narrative filter** (what just happened, what the relationship is, what the NPC "knows").

- Best use of FER is to **slightly adjust** facial expression, tone, or wording, not to dramatically flip the whole character mood each frame.
- For a dating / relationship-heavy scenario, emotional responsiveness matters more than raw accuracy; believable nuance wins over perfect classification. University of Twente Research

Takeaways

- Supports using FER-driven **affective mirroring** as part of the system, but with smoothing and context checks.
- Good precedent for small-N pilot studies testing emotional AI in games.